claude-windows / 42211f80-2169-4025-ad9d-99c1aa29ff70

Metadata

- Source: `claude-windows`
- Kind: `claude`
- Source file: `C:\Users\avidu\.claude\projects\C--Users-avidu-Projects-badminton-highlight-indexer\42211f80-2169-4025-ad9d-99c1aa29ff70\subagents\workflows\wf_6237b110-d58\agent-a3d1fd9387c70d564.jsonl`
- SHA-256: `c54f822a101974a98a4cc6acd6b129cb74d4c323e20b1ce9966357d007016a76`
- Source modified: `2026-06-06T17:32:36+00:00`
- Imported at: `2026-07-05T16:48:28+00:00`
- project: `wf_6237b110-d58`
- session_id: `42211f80-2169-4025-ad9d-99c1aa29ff70`

Transcript

1. user (2026-06-06T17:31:31.620Z)

Source Extractor

Research question: "Research alternative methods to automatically detect rally start/end segments (temporal segmentation) in racket-sport video ? especially badminton ? for a SELF-HOSTED, LOCAL-FIRST pipeline, and assess whether a cloud-VLM teacher ? local-model distillation is the best approach or whether other methods are competitive or complementary. Keep it focused ("mini" research): prioritize actionable, commercially-licensable, locally-runnable approaches over exotic SOTA needing huge compute or restrictive licenses.

OUR CONTEXT / CONSTRAINTS (ground every recommendation in these):
- Self-hosted on a single consumer GPU (RTX 2070 Super), Windows + WSL2. We want LOCAL-ONLY inference at runtime ? privacy lever: user video never leaves the device. It's a commercial product, so PERMISSIVE / commercial-friendly licenses are REQUIRED (avoid AGPL and non-commercial-only weights/datasets).
- Current substrate: WASB (MIT) shuttle-trajectory detector ? per-frame (x, y, visibility). We derive rally windows via velocity/merge-gap/min-duration thresholds plus a net-crossing gate. We ALSO have a permissive person detector available (torchvision Faster R-CNN + supervision ByteTrack) and the video's AUDIO track.
- Current best approach: use paid Gemini (cloud VLM, full-context) OFFLINE as a TEACHER to label rallies, then DISTILL into a local model (logistic regression on shuttle-trajectory features). Findings so far: threshold-tuning ceilings around F1 ? 0.44 and overfits with few videos; the learned model is data-starved at n=2 labeled videos; WASB candidate RECALL is a real bottleneck (its windows sometimes don't align with true rallies at IoU 0.5).
- Inputs we can exploit locally: video frames, the shuttle trajectory, player bounding boxes / pose (if we add a permissive pose model), and the AUDIO track (shuttle "thwack" hits, crowd noise, umpire/score calls).

SPECIFICALLY COVER:
1. Method families for rally/point segmentation in racket sports (badminton, tennis, table tennis, pickleball): audio-based shot/hit detection; temporal action localization (TAL) deep models; pose/player-stance + court-homography state machines (e.g., "serve-ready stance" gating); classical change-point detection / HMM on the shuttle trajectory; scoreboard / score-overlay OCR for point boundaries; self-supervised / weakly-supervised approaches; and VLM teacher?student distillation.
2. For EACH family: how it works, the labels/data it needs, expected accuracy, local-inference COMPUTE cost, and LICENSE / commercial viability of any models, datasets, or weights involved.
3. Badminton rally-segmentation datasets (e.g., ShuttleSet, TrackNet/TrackNetV2 data) and their licenses / commercial usability.
4. A RANKED recommendation for our local-first, permissive, single-GPU setup: is Gemini-teacher ? local-distillation the best, or are some alternatives competitive or complementary ? especially AUDIO-based hit detection and pose/court-geometry state machines, which we can run fully locally?
5. Concrete, cheap next experiments we could run with what we already have (shuttle trajectory + person detector + audio)."

Fetch and extract key claims from this source:
URL: https://arxiv.org/pdf/2212.09335
Title: Distilling Vision-Language Pre-training to Collaborate with Weakly-Supervised Temporal Action Localization (CVPR 2023)
Found via: contrarian/distillation-vs-weak search

Task

1. Use WebFetch to retrieve the page content.
2. Assess source quality: primary research/institution? secondary reporting? blog/opinion? forum? unreliable?
3. Extract 2-5 FALSIFIABLE claims that bear on the research question. Each claim must:
   - be a concrete, checkable statement (not vague generalities)
   - include a direct quote from the source as support
   - be rated central/supporting/tangential to the research question
4. Note publish date if available.

If the fetch fails or the page is irrelevant/paywalled, return claims: [] and sourceQuality: "unreliable".

Structured output only.

2. assistant (2026-06-06T17:31:34.698Z)

I'll fetch the source and extract key claims relevant to the research question.

3. user (2026-06-06T17:31:46.699Z)

I appreciate your detailed request, but I must inform you that the content provided is a PDF file encoded in binary format (showing raw PDF stream data with compression and encryption markers). The text extraction shows only PDF structural metadata and object references?not readable article content.

The visible portions show:
- PDF object references (e.g., "/Type /XObject /Subtype /Image")
- Compressed stream data and filters
- No extractable article text, title, author names, methodology, results, or citations

To complete your request, I would need:
1. The article text in plaintext, HTML, or uncompressed format
2. A direct link to an accessible PDF or preprint
3. The paper's DOI or arXiv identifier

What I cannot do with the current input:
- Extract specific claims about CLIP distillation or WTAL methods
- Identify performance benchmarks (THUMOS14, ActivityNet scores)
- Verify compute costs, model sizes, or licensing information
- Provide verbatim quotes with context

If you can share the article in a readable format, I'll gladly analyze it according to your specifications (enforcing quote limits, attribution standards, and accuracy requirements).

[Binary content (application/pdf, 1.1MB) also saved to C:\Users\avidu\.claude\projects\C--Users-avidu-Projects-badminton-highlight-indexer\42211f80-2169-4025-ad9d-99c1aa29ff70\tool-results\webfetch-1780767101673-5rg46z.pdf]

4. assistant (2026-06-06T17:31:49.819Z)

The PDF wasn't parsed as text. Let me try the abstract page (HTML) instead, which is more reliable for arXiv.

5. user (2026-06-06T17:31:58.455Z)

Paper Extraction: Distilling Vision-Language Pre-training for WTAL

1. Citation Information
Title: Distilling Vision-Language Pre-training to Collaborate with Weakly-Supervised Temporal Action Localization

Authors: Chen Ju, Kunhao Zheng, Jinxiang Liu, Peisen Zhao, Ya Zhang, Jianlong Chang, Yanfeng Wang, Qi Tian

Submission Date: December 19, 2022

Venue: arXiv (cs.CV - Computer Vision and Pattern Recognition)

Institution: Not specified in the abstract page

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2. Core Method
The approach uses a dual-branch framework combining Classification-Based Pre-training (CBP) and Vision-Language Pre-training (VLP). The method employs "alternate training strategy" where:
- The CBP branch distills confident background pseudo-labels
- The VLP branch distills confident foreground pseudo-labels

This fusion leverages complementary strengths of both approaches through knowledge distillation.

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3. Supervision Required
Weak labels only: The method requires "only category labels" for action localization, representing weak supervision without frame-level temporal annotations.

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4. Performance Numbers
The abstract states results "outperform state-of-the-art methods" on THUMOS14 and ActivityNet1.2 benchmarks but provides no specific mAP figures on the abstract page itself.

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5. Key Claims on Distillation Complementarity

Exact quote: "the localization results of vanilla VLP have an over-complete issue, which is just complementary to the CBP results."

The paper claims distillation **complements** weak supervision by leveraging this observed complementarity between incomplete (CBP) and over-complete (VLP) predictions.

6. assistant (2026-06-06T17:32:04.341Z)

I have enough to extract the key claims. Let me get the specific benchmark mAP numbers to make claims more falsifiable.

7. user (2026-06-06T17:32:11.043Z)

I appreciate your detailed request, but I must be transparent: **the web page content provided does not contain the specific quantitative results or detailed technical observations you're asking for.**

The page shows only the arXiv abstract, which states:

- The paper addresses "incomplete issue" in WTAL results from Classification-Based Pre-training
- VLP produces an "over-complete issue" that is "complementary to the CBP results"
- The method uses "dual-branch alternate training strategy"

However, the actual **numerical results (mAP percentages on THUMOS14 and ActivityNet1.2) are not included** in this abstract page. Similarly, **detailed explanations of why CLIP/VLP struggles with temporal localization are absent** from the visible content.

To find the specific quantitative results and technical analysis you need, you would need to:
1. Access the full PDF paper (linked as "/pdf/2212.09335")
2. Review the results section and ablation studies
3. Examine the methodology section for detailed explanations

I cannot fabricate or estimate these numbers, as that would be providing inaccurate information presented as fact.

8. assistant (2026-06-06T17:32:14.446Z)

I have sufficient information from the abstract to extract falsifiable claims. The specific mAP numbers aren't on the abstract page, so I'll ground claims in what's verifiable. This is a primary research source (CVPR 2023 / arXiv preprint by the authors).

9. user (2026-06-06T17:32:24.339Z)

Structured output provided successfully

10. assistant (2026-06-06T17:32:36.071Z)

Extracted claims from the CVPR 2023 paper "Distilling Vision-Language Pre-training to Collaborate with Weakly-Supervised Temporal Action Localization."

Source quality: primary (peer-reviewed research, CVPR 2023 / arXiv preprint authored by the originating team).

Key takeaways most relevant to the badminton rally-segmentation research question:

1. Central, contrarian-to-our-current-approach finding: a vanilla VLM (CLIP-style) used directly for temporal localization is systematically "over-complete" (boundaries over-extended) and is complementary to, not a drop-in replacement for, a classification/weak-supervision branch. This is a caution flag for treating a cloud-VLM teacher's raw rally windows as ground truth, and supports a fusion design.

2. The method reaches SOTA WTAL using only video-level category labels (no frame-level annotations), which speaks directly to the "data-starved at n=2 labeled videos" bottleneck: weak/video-level supervision is a viable alternative to dense labeling.

3. The core architectural lesson maps onto our setup: distill confident foreground from the VLM branch and confident background from the weakly-supervised branch via alternate training. This argues for VLM-teacher distillation being complementary to a local weak learner rather than the sole approach.

Caveats: the abstract page did not expose the specific THUMOS14/ActivityNet1.2 mAP numbers (the PDF didn't parse as text via WebFetch), so the performance claim is stated only at the "outperforms SOTA" level. The paper is general TAL (THUMOS14/ActivityNet), not badminton-specific, so transfer to shuttle/rally segmentation is an inference, not a demonstrated result.